

Week 03 — Backshift Operator, Wold Decomposition, Causality & Invertibility

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The big picture. In polynomial form $\Phi(B)X_t = \Theta(B)\varepsilon_t$, the location of the roots relative to the unit circle decides stationarity (causality) and invertibility.

This week we repackage AR, MA and ARMA models with the **backshift operator** B . Difference equations become an algebra of *polynomials* $\Phi(z), \Theta(z)$, and every structural question (Is it stationary? Invertible? What is its MA(∞) form?) turns into a question about the *roots* of these polynomials. Throughout, $\{\varepsilon_t\}$ is mean-zero white noise with $\text{Var}(\varepsilon_t) = \sigma^2$.

Key definitions

Definition — Backshift operator B

B shifts a series back one step in time: $BX_t = X_{t-1}$, and iterating $B^k X_t = X_{t-k}$ (with $B^0 = I$). It is linear and shifts commute, so polynomials in B multiply and reorder like ordinary polynomials.

Why treat B as a variable z ?

B is just “ $X_t \mapsto X_{t-1}$ ”. But treating it as an algebraic symbol z lets us *factor, invert* and *expand* model equations. Every result this week is really a statement about the polynomial $\Phi(z)$ or $\Theta(z)$ you get by replacing B with the complex variable z .

Definition — ARMA in polynomial (backshift) form

An ARMA(p, q) process is written compactly as

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \Phi(z) = \sum_{j=0}^p \phi_j z^j, \quad \Theta(z) = \sum_{k=0}^q \theta_k z^k.$$

$\Phi(z)$ is the **characteristic polynomial of the AR part**, $\Theta(z)$ that of the **MA part**. Pure cases: AR(p) is $\Phi(B)X_t = \varepsilon_t$; MA(q) is $X_t = \Theta(B)\varepsilon_t$. The *roots* of these polynomials govern stationarity and invertibility.

Definition — General linear process (one-sided MA(∞))

A general linear process is

$$X_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 = \sum_k g_k^2 < \infty, \quad g_0 = 1.$$

It depends only on *past and present* noise, so it is **causal**. In operator form $X_t = G(B)\varepsilon_t$ with the **transfer function** $G(z) = \sum_{k \geq 0} g_k z^k$. (A *two-sided* linear process also allows $k < 0$, i.e. future noise.)

Definition — Transfer function: zeros and poles

For $X_t = G(B)\varepsilon_t$ with rational $G(z) = G_1(z)/G_2(z)$: roots of the numerator G_1 are the **zeros** of G ; roots of the denominator G_2 are the **poles**. For an ARMA, $G(z) = \Theta(z)/\Phi(z)$, so *roots of Φ are the poles and roots of Θ are the zeros*. Whether G expands using only past/present (vs. future) noise is decided by where these roots sit relative to $|z| = 1$.

The summability $\|g\|_2^2 < \infty$ is what makes the moments finite: $\mathbb{E}[X_t] = 0$, $\text{Var}(X_t) = \sigma^2 \|g\|_2^2 < \infty$, so any linear process is second-order stationary. An **innovations outlier** is a value of ε_t that affects X_t and all subsequent $X_{t+1}, X_{t+2}, \dots$ — exactly the forward-propagation of a one-sided process.

Key results & formulas

Theorem — Wold decomposition

Any zero-mean second-order stationary $\{X_t\}$ splits as

$$X_t = U_t + V_t, \quad U_t \perp V_t,$$

where $U_t = \sum_{k \geq 0} g_k \varepsilon_{t-k}$ is a one-sided MA(∞) ($g_0 = 1$, $\|g\|_2^2 < \infty$, $\{g_k\}, \{\varepsilon_t\}$ *uniquely* determined), and V_t is **singular** (deterministic: perfectly predictable from its own past). *Idea*: project X_t onto the span of its own past; the one-step residuals are the white noise ε_t , the g_k are projection coefficients, V_t is the infinite-past part.

What Wold really says

The only *new information* arriving at time t is the innovation ε_t ; everything else is either an echo of past innovations (U_t) or perfectly forecastable (V_t , e.g. a fixed sinusoid). For the processes we study $V_t = 0$, leaving the pure MA(∞) form $X_t = G(B)\varepsilon_t$. This is *why* the linear/ARMA world is central: ARMA models are finite rational approximations to this MA(∞).

ACVS of a causal linear process. Once the MA(∞) weights $\{g_k\} = \{\psi_k\}$ are known,

$$\gamma_\tau = \sigma^2 \sum_{k=0}^{\infty} g_k g_{k+|\tau|}.$$

This is the engine for the autocovariance of *any* causal ARMA.

The root test (causality & invertibility)

For $\Phi(B)X_t = \Theta(B)\varepsilon_t$:

AR(p) : stationary $\iff$ roots of Φ outside $|z| = 1$; *always* invertible.

MA(q) : *always* stationary; invertible $\iff$ roots of Θ outside $|z| = 1$.

ARMA(p, q) : stationary $\iff$ roots of Φ outside; invertible $\iff$ roots of Θ outside.

One test, two polynomials

Stationarity (causality) and invertibility are the *same* test (“all roots outside the unit circle”) applied to *two different* polynomials. “Roots of Φ outside” means $X_t = G(B)\varepsilon_t$ uses only past/present *noise* (causal). “Roots of Θ outside” means $\varepsilon_t = G^{-1}(B)X_t$ uses only past/present *data* (invertible). An AR has no Θ to fail $\Rightarrow$ always invertible; an MA has no Φ to fail $\Rightarrow$ always stationary.

Why “outside”. For $X_t = G(B)\varepsilon_t$, the one-sided (past/present) expansion of $1/G_2$ exists *iff* all poles of G lie outside $|z| = 1$ — equivalently $|G(z)| < \infty$ for $|z| \leq 1$, i.e. G is *analytic inside and on* the unit circle. The Laurent expansion shows a pole *inside* the circle forces a *future-noise* branch.

Reciprocal-root warning

“Outside” refers to roots of $\Phi(z) = \sum_j \phi_j z^j$, i.e. $|z| > 1$. In terms of *reciprocal* roots $g = 1/z$ this is $|g| < 1$ — exactly the AR(1) condition $|\phi| < 1$ (root of $1 - \phi z$ is $z = 1/\phi$). Do not mix the conventions, and never check *inside* the circle.

Diagnostic mirror. A causal AR(p) has $\alpha_\tau = 0$ for $\tau > p$ (PACF cuts off at lag p) — the mirror image of the ACF cutting off at lag q for an MA(q).

ψ -weights by coefficient matching. For a causal ARMA, equate $\Phi(z)(\sum_{j \geq 0} \psi_j z^j) = \Theta(z)$ and match powers of z , giving the forward recurrence $\psi_j = \theta_j^* + \sum_{i=1}^p \phi_i \psi_{j-i}$ ($\psi_0 = 1$). The closed form is a sum $\sum_i (\text{poly in } j) r_i^{-j}$ over the AR roots r_i .

Spectral density from the polynomials. For a causal, invertible ARMA,

$$S(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2}, \quad -\frac{1}{2} \leq f \leq \frac{1}{2}.$$

Intuition

Tiny worked example — the root test on an ARMA(2,1)

For $Y_t - \frac{1}{4}Y_{t-1} + \frac{1}{16}Y_{t-2} = \varepsilon_t - \frac{1}{2}\varepsilon_{t-1}$: write $\Phi(z) = 1 - \frac{1}{4}z + \frac{1}{16}z^2$ and $\Theta(z) = 1 - \frac{1}{2}z$.

Stationary? $\Phi(z) = 0 \iff z^2 - 4z + 16 = 0 \iff z = 2 \pm 2\sqrt{3}i$, $|z| = \sqrt{16} = 4 > 1 \checkmark$

Invertible? $\Theta(z) = 0 \iff z = 2$, $|2| = 2 > 1 \checkmark$

All roots outside the unit circle, so the process is both stationary and invertible.

Key takeaways

Key takeaways — the week’s reflexes

- **Translate first:** put the model in backshift form $\Phi(B)X_t = \Theta(B)\varepsilon_t$ and read off $\Phi(z), \Theta(z)$.
- **Root test:** stationary $\iff$ all roots of Φ have $|z| > 1$; invertible $\iff$ all roots of Θ have $|z| > 1$. Two independent checks.
- **Free checks:** never test an AR for invertibility (always invertible) nor an MA for stationarity (always stationary).
- **Find the MA(∞) weights ψ_j** by matching $\Phi(z) \sum \psi_j z^j = \Theta(z)$, then get the ACVS via $\gamma_\tau = \sigma^2 \sum_k \psi_k \psi_{k+|\tau|}$.
- **Wold** = every stationary process is (a deterministic part plus) a one-sided MA of white noise; the innovation ε_t is the only new information at time t .
- **Convention:** “roots outside $|z| = 1$ ” $\equiv$ “reciprocal roots inside” ($|\phi| < 1$ for AR(1)). Don’t check inside the circle.